

Backend Developer Task (Python/Django Focused)

Task: Background Job Based Notification System

Build a backend API where users can schedule notifications.

Requirements

Use:

- Django + Django REST Framework
- PostgreSQL
- Docker

Features

Create APIs for:

- User authentication
- Create notification
- Schedule notification
- View notification history
- Retry failed notification

Notification Object

Each notification contains:

- Title
- Message
- Scheduled Time
- Status
- Retry Count

Critical Thinking Requirements

Candidate must handle:

- Time validation
- Failed job handling
- Retry logic

- Background processing
- API security
- Edge cases
- Queue/job architecture thinking

Extra Problem-Solving Condition

If scheduled time is in the past:

- API should reject the request

If notification fails 3 times:

- Mark as permanently failed
- Prevent infinite retries

Bonus

- Celery/Redis integration
- Docker Compose setup
- Logging & monitoring

Evaluation Focus

- Backend architecture
- Queue logic understanding
- Security
- Scalability thinking
- Database design
- Clean API structure
- Error handling quality